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Time : 3.00 hrs.

Quarterly Examination - 2026

MATHEMATICS

Reg. No. | | | | | | | |

Max. Marks : 90

PART – I

Note : (i) All questions are compulsory. (ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

1. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4x - 2\sin^2x + 1$ is a) 2 b) 4 c) 1 d) ∞

2. The product of all four values of $\left(\cos\frac{\pi}{3} + 1\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is a) -2 b) -3 c) 1 d) 2

3. The principal argument of $\frac{3}{-1+i}$ is a) $\frac{-5\pi}{6}$ b) $\frac{-2\pi}{3}$ c) $\frac{-3\pi}{4}$ d) $\frac{-\pi}{2}$

4. If the directrix cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < c < 1$

5. If α, β and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum\frac{1}{\alpha}$ is a) $\frac{-q}{r}$ b) $\frac{-p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$

6. If $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, B = adjA and C = 3A, then $\frac{|\text{adj}B|}{|C|}$ = a) $\frac{1}{3}$ b) $\frac{1}{9}$ c) $\frac{1}{4}$ d) 1

7. $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{c} = 3\hat{i} + 5\hat{j} - \hat{k}$ then a vector perpendicualr to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is a) $-17\hat{i} + 21\hat{j} - 97\hat{k}$ b) $-17\hat{i} + 21\hat{j} - 123\hat{k}$ c) $-17\hat{i} - 21\hat{j} + 97\hat{k}$ d) $-17\hat{i} - 21\hat{j} - 97\hat{k}$

8. If $|x| \le 1$, then $2\tan^{-1}x - \sin^{-1}\frac{2x}{1+x^2}$ is equal to a) $\tan^{-1}x$ b) $\sin^{-1}x$ c) 0 d) π

9. The area of quadrilateral formed with foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is a) $4(a^2 + b^2)$ b) $2(a^2 + b^2)$ c) $a^2 + b^2$ d) $(a^2 + b^2)$

10. The point of contact of parabola is a) $c = \frac{a}{m}$ b) $\left(\frac{-a^2m}{c}, \frac{b^2}{c}\right)$ c) $\left(\frac{-a^2m}{c}, \frac{-b^2}{c}\right)$ d) $a = \frac{c}{m}$

11. $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is a) 17 b) 14 c) 19 d) 21

12. If $[\vec{a}, \vec{b}, \vec{c}] = 1$, then the value of $\frac{\vec{a}.(\vec{b}\times\vec{c})}{(\vec{c}\times\vec{a}).\vec{b}} + \frac{\vec{b}.(\vec{c}\times\vec{a})}{(\vec{a}\times\vec{b}).\vec{c}} + \frac{\vec{c}.(\vec{a}\times\vec{b})}{(\vec{c}\times\vec{b}).\vec{a}}$ is a) 1 b) -1 c) 2 d) 3

13. Let $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and $4B = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & x \\ -1 & 1 & 3 \end{bmatrix}$. If B is the inverse of A, then the value of x is a) 2 b) 4 c) 3 d) 1

14. If $|z| = 1$, then the value of $\frac{1+z}{1+\bar{z}}$ is a) z b) $\bar{z}$ c) $\frac{1}{z}$ d) 1

15. The arg of i is a) 0 b) 1 c) -1 d) Not defined

16. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (11, 2) the coordinates of the other end are a) $(-5, 2)$ b) $(-3, 2)$ c) $(5, -2)$ d) $(-2, 5)$

17. If $\cot^{-1}x = \frac{2\pi}{5}$ for some $x \in R$, the value of $\tan^{-1}x$ is a) $\frac{-\pi}{10}$ b) $\frac{\pi}{5}$ c) $\frac{\pi}{10}$ d) $\frac{-\pi}{5}$

18. An ellipse has OB as semi minor axes, F and F^1 its foci and the angle FBF1 is a right angle. Then the eccentricity of the ellipse is a) $\frac{1}{\sqrt{2}}$ b) $\frac{1}{2}$ c) $\frac{1}{4}$ d) $\frac{1}{\sqrt{3}}$

19. If $\sin^{-1}\frac{x}{5} + \text{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is a) 4 b) 5 c) 2 d) 3

20. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

PART - II

Note : i) Answer any seven questions. ii) Q.No.30 is compulsory. **7 x 2 = 14**

21. Find $\tan(\tan^{-1}(2019))$

22. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solution.

23. Which one of the points $10 - 8i$, $11 + 8i$ is closest to $1 + i$.

24. Find the matrix A if $\text{adj}(A) = \begin{bmatrix} 7 & 7 & -7 \\ -1 & 11 & 7 \\ 11 & 5 & 7 \end{bmatrix}$

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25. Find the distance between the parallel planes $x + 2y - 2z + 1 = 0$ and $2x + 4y - 4x + 5 = 0$
26. If $y = 4x + c$ is a tangent to the circle $x^2 + y^2 = 9$, find C.
27. Solve the equation $x^3 - 3x^2 - 33x + 35 = 0$
28. Find the rank of the matrix $\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$ by minor method.
29. Find the value of $\sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right)$
30. Find the value of $\sum_{k=1}^{8}\left(\cos\frac{2k\pi}{9} + 1\sin\frac{2k\pi}{9}\right)$

PART - III

Note : i) Answer any seven questions. ii) Q.No.40 is compulsory. **7 x 3 = 21**

31. Find the principal argument Arg z, when $z = \frac{-2}{1+i\sqrt{3}}$
32. Show that the equation $|3z - 6 + 12i| = 8$ represent a circle, and find its center and radius.
33. The parabolic communication antenna has a focus at 2m distance from the vertex of the antenna. Find the width of the antenna 3m from the vertex.
34. Find the acute angle between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 4\hat{k}) + t(2\hat{i} + 2\hat{j} + \hat{k})$ and the straight line passing through the points (5, 1, 4) and (9, 2, 12)
35. $2x + 3y - z = 9$, $x + y + z = 9$, $3x - y - z = -1$, solve the system of linear equations by matrix inversion method.
36. If k is real, discuss the nature of the roots of the following polynomial equation $2x^2 + kx + k = 0$, in terms of k.
37. If the normal at the point 't_1' on the parabola $y^2 = 4ax$ meets the parabola again at the point 't_2' then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$
38. Find the vector and cartesian equations of the plane passing through the point with position vector $2\hat{i} + 6\hat{j} + 3\hat{k}$ and normal to the vector $\hat{i} + 3\hat{j} + 5\hat{k}$.
39. Find the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$
40. If $\text{adj}A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ and $\text{adj}B = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$ then adj(AB) is?

PART - IV

Note : i) All questions are compulsory. **7 x 5 = 35**

41. a) By using Gaussian elimination method, balance the chemical reaction equation. $C_5H_8 + O_2 \rightarrow CO_2 + H_2O$ **(OR)**

    b) If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}$, $\vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$ verify that $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c}$
42. a) Solve the equation $z^3 + 8i = 0$, where $z \in C$ **(OR)**

    b) Two coast guard stations are located 600km apart at points A(0, 0) and B(0, 600). A distress signal from a ship at P is received at slightly different times by two stations. It is determined that the ship is 200 km farther from station A than it is from station B. Determine the equation of hyperbola that passes through the location of the ship.
43. a) If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation $x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x - 140 = 0$, find all roots. **(OR)**

    b) Show that the lines $\frac{x-3}{3} = \frac{y-3}{-1}$, $z - 1 = 0$ and $\frac{x-6}{2} = \frac{z-1}{3}$, $y - 2 = 0$ intersect. Also find the point of intersection.
44. a) Evaluate $\sin\left[\sin^{-1}\left(\frac{3}{5}\right) + \sec^{-1}\left(\frac{5}{4}\right)\right]$ **(OR)**

    b) Show that the points $1, \frac{-1}{2} + i\frac{\sqrt{3}}{2}$ and $\frac{-1}{2} - i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle.
45. a) Find the domain of $\sin^{-1}(2 - 3x^2)$ **(OR)**

    b) Find the equation of the circle passing through the points (1, 1), (2, –1) and (3, 2)
46. a) Solve the following equation. $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$ **(OR)**

    b) $2x + 4y + 6z = 22$, $3x + 8y + 5z = 27$, $-x + y + 2z = 2$ solve the given system of linear equations by Gaussian elimination method.
47. a) Find parametric form of vector equation and cartesian equation of the plane passing through the points (2, 2, 1), (1, –2, 3) and parallel to the straight line passing through the points (2, 1, –3) and (–1, 5, 8) **(OR)**

    b) Find the equation of the ellipse whose length of latus rectum 4, distance between foci $4\sqrt{2}$, center (0, 0) and major axis as y-axis.

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